Decimal Increments

1/16 in	0.0625
1/8 in	0.1250
3/16 in	0.1875
1/4 in	0.2500
5/16 in	0.3125
3/8 in	0.3750
7/16 in	0.4375
1/2 in	0.5000
9/16 in	0.5625
5/8 in	0.6250
11/16 in	0.6875
3/4 in	0.7500
13/16 in	0.8125
7/8 in	0.8750
15/16 in	0.9375
1 in	1.0000

Pipe Wall Thickness (I.P.)

Call Out Size	Wall Thickness
1/2-2 1/2 inch	3/8 inch Total
3-5 inch	1/2 inch total
6-8 inch	5/8 inch total
10-12 inch	3/4 inch total

Note: Above 12 inch, the callout size is the actual O.D. of the pipe.

Math

Mitres:

Mean Radius ±1.2 O.D X 1/2 π (pi) ÷ Miters

Lags:

O.D. ÷ I.D. X ILS = (OLS)

ID ÷ OD X OLS = (ILS)

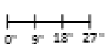
Head Segments:

See width X R1 –7 ÷ R1= Seg Widths

Line Division

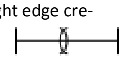
Mathematically:

Use easy math when possible.



Basic Geometry:

Using dividers set more than half the line length. From each end of the line scribe inward creating common apex's above and below the line. Join the apex's with a straight edge creating a center point.



Ruler Method:

Draw a line on one end of the horizontal line at 90°. Using a rule place one end of the horizontal line. Slide the other end up the vertical line until you reach a number divisible by the number of divisions required.

Trial & Error:

Guessing—Set Dividers—Step off required number of divisions. Divide the remainder by the number of divisions. Add this number to the divider, step off again.

Circle Division (Bisecting)

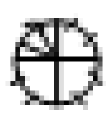
Important!!

Draw cross hairs then add circle.

Place dividers on quarter point set to just over half a 1/4 point and scribe outside the circle slightly. Repeat for next 1/4 point joining arcs for a common apex. Repeat as required.

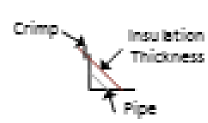
Bisecting by thirds:

After scribing circle keep radius setting. Scribe for 1/4 points both ways onto the circle.



Bevels: Square Method

1. Draw a 90° angle large enough to fit the OD of the pipe covering. (Legs)
2. Mark pipe radius on both legs.
3. Mark Insulation radius on both legs. Connect matching radii. Extend line for insulation radii, outward to allow for crimp.
4. Set dividers from insulation thickness -> crimp allowance. Then scribe a circle.
5. Repeat for insulation radii. Then pip radii—using same center point for all 3 circles.
6. Cut out and fabricate removing less than 1/4 for closure when applying.



Equal Tee

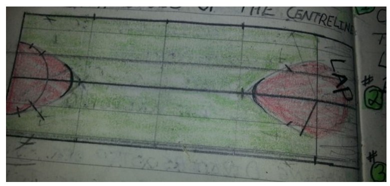
Dims Required =OD&OC Insulation. OC÷4

Tee Portion:

1. Cut a piece of metal equal to the OC+ LAP and ÷ into 4 equal parts (excluding LAP).
2. Set dividers to radius and scribe an arc in the lower corner. Then locate *A* 1/4 from the top of the arc.
3. Locate 1/8" *B* from 1/4 OC and mark
4. Using dividers step off 3 Radii. Use this setting to find the common apex then join *A* and *B*.

Body Portion:

1. Cut a piece of metal equal to the OC plus LAP
2. Locate the center of the cut-out. Repeat "T" portion on both sides of the centre line.

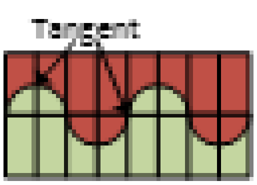


Unequal Tee

Dims Required: OD & OC of both pipes (Insulation included) & amount of change (depth of cheek)

Tee Portion

1. Cut a piece of metal equal to the OC of the small pipe + Lap, ÷ into equal number of parts (excluding LAP).
2. In the lower corner scribe an arc = the radius of the larger pipe
3. From the bottom of the arc, scribe a perpendicular line equal to the small radius.
4. Square the small radius and measure the depth of cheek. Transfer this measurement to the underside of the arc.
5. Draw a line across the metal indicating (depth of cheek).
6. Using the big radius alternate arc up and down at each 1/4 point.
7. Connect the arcs at the tangents using a ruler, flare if needed.

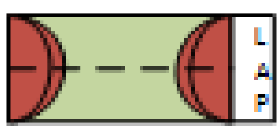


Unequal Tee

Dims Required: OD & OC of both pipes (Insulation included) & amount of change (depth of cheek)

Body Portion

1. Cut a piece of metal equal to the large OC + LAP. Make a center line for the cut out.
2. Set dividers to the small radius and scribe an arc on both sides of the metal for the cut-out. (Use the center line as the divider point).
3. Move dividers into the metal 1/8 and re-scribe both sides.



Equal Laterals

Dims Required: OD of pipe (Ins. included)

Working Drawing

1. Draw the working drawing. Draw center lines, add OD of pipe and scribe arcs on all ends. Divide arcs equally and project lines into drawing and connect.

Tee Portion

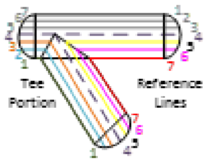
1. Cut a piece of metal equal to the OC + LAP and ÷ into same number of parts as the working drawing and label.
2. Transfer dimensions from the working drawing to the stretch out.

Body Portion

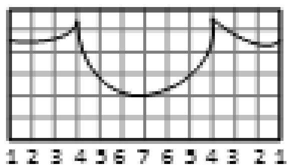
1. Cut a piece of metal equal to the OC + LAP ÷ into same number of equal portions as the tee portion. Number according to LAP location.
2. Transfer points from body portion of the working drawing. #5 goes on line #3, #6 on #2, #7 on #1.
3. Take all dims from one side of the working drawing.

Equal Laterals: Drawings only

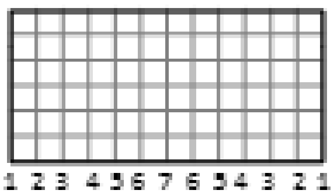
Working Drawing



Tee Portion



Body Portion



Unequal Laterals

Required Dims: OD Both pipes, OC both pipes (Ins included).

Working Drawing

1. Draw the working drawing. Start with the centerline of the large pipe (Top portion not required).
2. Draw a secondary projection downward form the body portion and divide, carry division lines up to the body portion connecting at the arc, then across and down the pre-divided tee portion.

Tee Portion

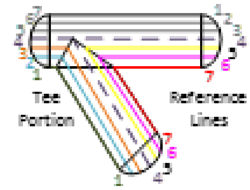
1. Cut a piece of metal equal to the OC of the small pipe + LAP and Divide.
2. Using dividers transfer small pipe reference points to the stretch-out.

Body Portion

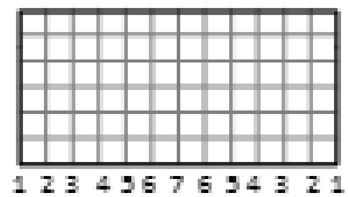
1. Cut a piece of metal equal to the large OC + LAP
2. Set dividers to along the large arc that were generated from the secondary projection. Transfer these individual widths into the circ of the metal.
3. Transfer the dimensions from the body portion of the working drawing. Take all dims from one side. #5 goes on line #3, #6 on #2, and #7 on #1.

Unequal Laterals

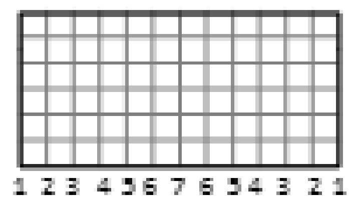
Working Drawing



Tee Portion



Body Portion



Calculating Elbows

Mean Radius MR
Long Radius 1 1/2 times callout size
Short Radius equal to call out size (as measured from the apex of the two welds.
Formulas: OD includes insulation
Heel: (MR+1/2 OD)*1/2 Pi ÷ Number of Mitres
(12X1 1/2 LR)(18-8=10)X1.57÷10=1.57"

Gore Elbows

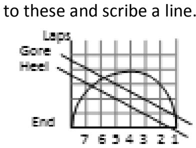
Dims Required OD, OC, Heel & Throat Measurements, Side & End Laps

Working Drawing

Cut a piece of metal equal to the OC + Total End Lap

Scribe a semi circle equal to the OD of the insulation. Divide into equal parts and add lines vertically and label.

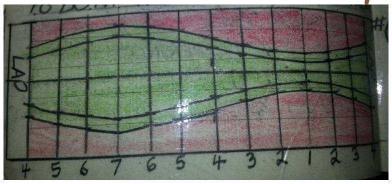
Place 1/2 the heel of 1 Gore on 1 end of the circle (#1 or #7) then place 1/2 the throat on of 16 gore on the line opposite the heel line, add 1/2 the total side up to these and scribe a line.



Stretch-Out

Cut a piece of metal equal to the OC+End Laps & Draw a center line along the OC length and divide into the same divisions as the working drawing.

Using dividers place one tip on the center line at the division intersections and scribe the corresponding working drawing dimensions to both sides of the centerline.



Compressed Gore Pattern

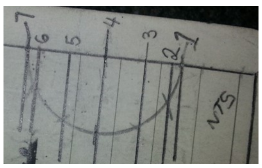
Scribe a vertical and horizontal lines large enough for the heel measurement to be placed on the vertical line.

Measuring up from the horizontal place 1/2 the heel measurement (of 1 gore) on the vertical line and 1/2 the throat on the vertical line also.

Find the center between the 1/2 heel and 1/2 throat and use dividers to join these with a semi circle.

Divide the semi circle into 6 equal divisions and label.

Transfer these points to the stretch-out as explained on the previous page.



Butterfly Gores

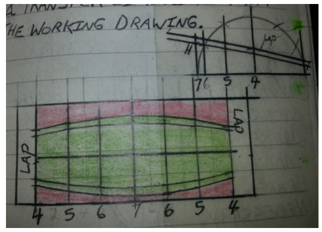
Dimas Required: OD, OC Heel, Throat, Total Side Lap, Total End Lap

Heel

Draw the working drawing

As with regular gore drawing cut metal to 1/2 OC plus laps

Divide into 6 equal parts & transfer dimensions from the working drawing.



Throat

Cut metal 1/2 OC + Lap & scribe cross hairs in the center.

Set dividers to radius and place 1 time in the center of crosshairs & scribe an arc on both sides of the centre of the metal.

Place 1/2 the total throat on each side of the center minimum 1/2" inch for band.

With dividers @ radius scribe arcs outward from the 1/2 throat marks.

Add laps and flares as necessary.

Drawing on page 28.

Concentric Reducer

Large OD Small OD Actual Height (Distance covered)

Radial Line Method

1. Using a piece of metal large enough for the amount of reduced to be done. Draw the side view.
2. Extend the sides up to a common apex.
3. Set your dividers from the apex to the top edge of the cone and scribe an arc.
4. Set your dividers from the apex to the bottom edge of the cone and scribe an arc.
5. Measure large OC and mark <u>along</u> the large arc line. Add LAP, cut away excess and assemble.

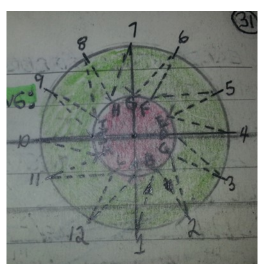
Triangulation

1. Draw the working drawing (End view)
2. Divide the large and small circles into equally spaced parts and label.
3. Draw the true length triangle. And mark the actual height.
4. Develop the true lengths and mark them on the triangle.
5. Develop the stretch-out start with

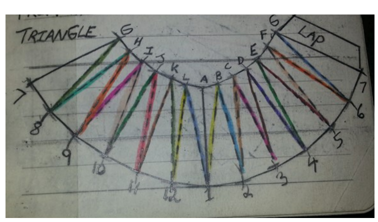
the slant height on the center line.

6. Develop the other points along the stretch-out.
7. Connect the dots and develop the pattern.
8. Add flares crimp allowance and LAP

Working Drawing



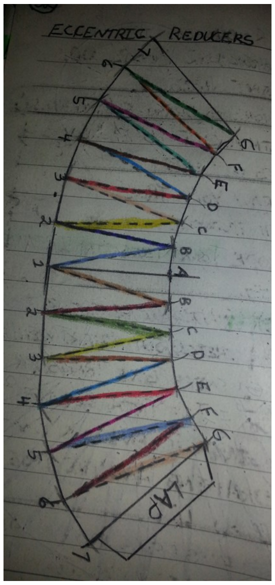
Stretch-out: Transfer dimensions from the working drawing & true length triangle.



Eccentric Reducers

Dims Required: Large OD, Small OD, Actual or Slant Height

1. Draw the working drawing (end view)
2. Divide both arcs into equally spaced parts and label.
3. Draw your true length triangle and develop your true lengths
4. Develop the stretch-out start with a center line, transfer the actual height , triangulate the reducer using dimensions on the working drawing and the true length triangle.
5. Connect points using a ruler, add flares and laps if needed, cut out and fabricate.



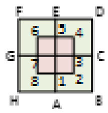
Square to Square

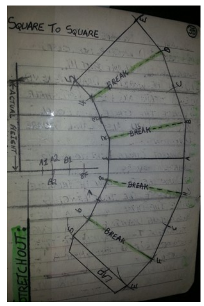
Dims Required: Top Dimension, bottom dimensions, slant height.

1. Draw the working drawing (end view).
2. Draw the true length triangle and develop the true lengths

HINT: To find the actual height, place the A1 dimension from the working drawing on the horizontal line of the true length triangle, set dividers to slant height and scribe across the vertical line.

3. Develop the stretch-out using triangulation (2 points to find the third).

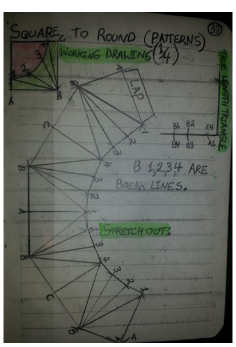




Square to Round

Dims Required: Diameter, Perimeter, Actual or slant height.

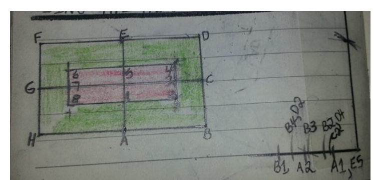
1. Draw the working drawing (end view)
2. Divide the circle into equal parts and label. Also label square dimensions
3. Connect circle divisions to square connectors.
4. Draw the true length triangle & develop the true lengths
5. Use the dimensions from the working drawing and true length triangle to triangulate the stretch out, start with a center line.
6. Add flares and laps as required. Cut out and assemble.

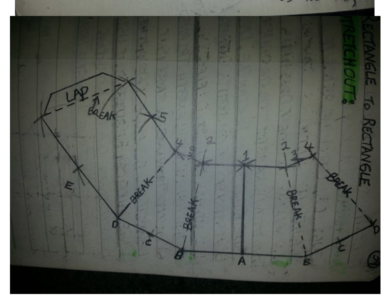


Rectangle to Rectangle

Dims Required: Large and small dims, actual or slant height.

1. Draw the working drawing (end view) include the cross hairs, divide & label (best to include the break lines).
2. Draw the true length triangle and develop the true lengths.
3. Use the dimensions on the working drawing and true length triangle to develop the stretch-out using the triangulation method.

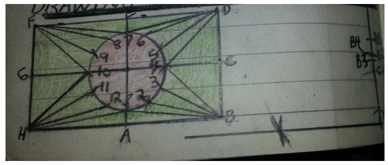


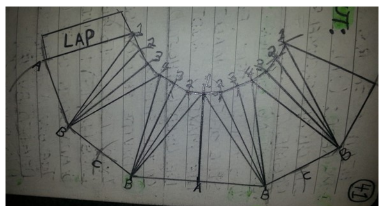


Rectangle to Round

Dims Required: Dimensions of rectangle, OD of round, actual or slant height.

1. Draw the working drawing (end view). Start with cross hairs the circle then rectangle. Label divisions and connect to corners for break lines.
2. Draw the true length triangle and develop the true lengths using the working drawing.
3. Develop the stretch-out using dimensions on the working drawing and true length triangle.





Head Segments

Dims Required: OD OC Segment length (1/4 Circumference), segment width, number of segments.

Calculating a Segment Width

1. Set a straight edge on the side of the head to determine the maximum width that can be used without deflecting on the sides.
2. Divide this measurement into the OC to get the number of lags (peels) required.
3. Adjust this number up or down to get a number that is easily divided by 4.
4. Redivide this number of segments into the circumference to get the actual segment width.

Quick Method:

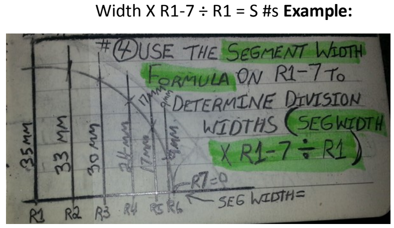
Divide the OC by 24, 28, 30, 32, 36 or 40. This will give you the seg width.

Mathematical Method: (Special Only)

1. Draw the side view of the head (1/4 circumference)
2. Divide the 1/4 arc into equally spaced parts and label.
3. Draw lines from the center line 90° through the arch @ each of the divisions & measure each line to the arc.
4. Use the segment width formula on R1-7 to determine division widths (seg width X R1 –7 ÷ R1).

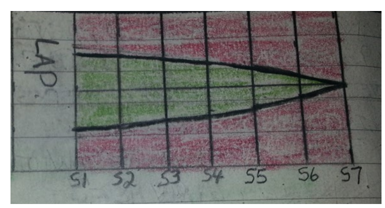
Head Segments: Stretch-out:

1. Cut a piece of metal 1/4 of the OC + Bottom lap.
2. Layout the segment length & divide into the same # of parts as the working drawing & label. Add a center line.
3. Mark 1/2 the measurements on each side of the center line. **Formula:** Seg Width X R1-7 ÷ R1 = S #s **Example:**



16^{mm} X $^{R3}30^{mm} \div 35^{mm} = 13.71^{mm}$

4. Add Lap as required and use as a pattern.



Head Segments (Chalk line method)

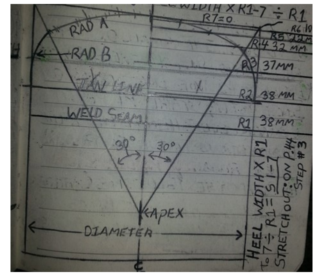
1. Mark a centerline on the head from tip to edge.
2. Mark 1/2 the segment width on each side of the center line.
3. Place a chalk line on seg width lines and extend to head tip & snap a line.
4. Divide the center line into equally spaced parts and measure the seg width @ each of these divisions.
5. Lay out the seg length (1/4 OC) & divide into the same # of divisions as the head center line.
6. Transfer the measurements form the head to the stretch-out, add laps and process.

Semi Elliptical Head Segments

Dims Required: OD & OC of head seg width, side & end laps, # of segments, seg length, weld to tangent (tan line).

Seg Width: Divide OC by an even # equally divisible by 4, eg. 40, 44, 48. These would be the number of segs. The resulting math is seg width.

1. Draw the working drawing (side view). Start with a center line then place 1/2 the diameter on each side. Use roughly 1/4 diameter to find the tan line (bottom of where the curve beins).
2. Divide into equally spaced parts and label.
3. Project lines from division marks to the center line and measure.
4. Use the seg width formula to determine the width @ each division. Heel width X R1 –7 ÷ R1

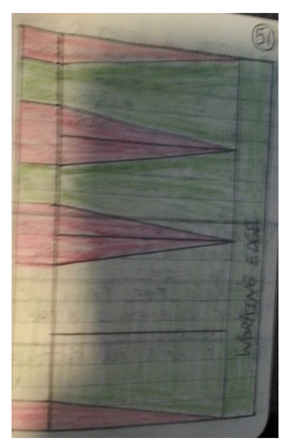


Concentric Armaflex Reducer

1. Cut a piece equal to the large OC (Insulation included) & the actual height + 2 working edges.
2. Divide into 4 equal parts & put a working edge on the top & bottom.
3. Subtract the small OC from the large OC & **divide by 4.**
4. Place 1/2 the dimension on each side of the 1/4 points.

Eccentric Armaflex Reducer

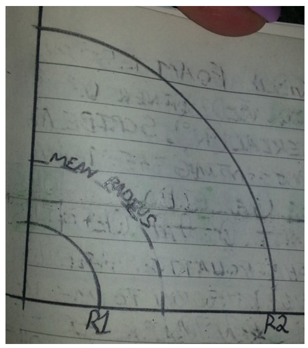
1. Cut a piece equal to the large OC (Insulation included) & the actual height + 2 working edges.
2. Divide into 4 equal parts & put a working edge on the top & bottom.
3. Subtract the small OC from the large OC & **divide by 3.**
4. Place 1/2 the dimension on the side of 4 of the 4 1/4 points.
5. Mark lines from the above dimensions to the 1/4 points they span.
6. Cut out and glue together



Extruded Foam Elbows

1. On a square corner of the material (90°) . Scribe an arc representing the mean radius of the elbow.
2. Subtract 1/2 the OD of the pipe (including wall thickness) from the mean radius then mark R1
3. Add 1/2 the OC Insulation included to R1 then put 1 tip of the divider on the apex and scribe R2
4. Cut out and use as a pattern.

Note: A chart may be used to get the dimensions required to layout an elbow.



Figuring Lag Sizes

Dims Required: ID of Insulation, OD of Insulation, Inside Lag size or outside lag size.

Formulas:

OD ÷ ID X ILS = OLS

ID ÷ OD X OLS = ILS

Set up the equation as follows, inside numbers on top.

$$\frac{I40}{O44} X \frac{2.727}{2.9997}$$

Always cross multiple then divide.

Finding the inside lag size:

1. Hold a ruler up the side of the tank, vessel or pipe to be insulated. (Image on page 54)
2. Looking down the rules note the measurement before the deflection on each side of where the rules touches is to much (roughly 1/8" gap). This measure is the ILS.

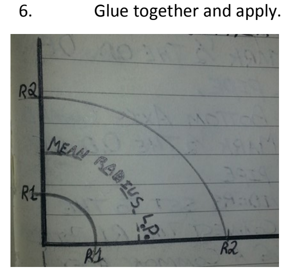
Tip: Consider the material being used. MF is less rigid than cal-sil, there fore due to compression of the material the MF lag size may be wirder (more delfection when measuring).

Extruded Foam Reducing Elbows

Eccentric: Always long radius.

Dims Required: OD & OC of SM & Lrg Pipes, Insul Thickness.

1. Scribe an arc equal to the mean radius of the large pipe.
2. On the bottom axis towards the paex mark 1/2 the OD of the large pipe & scribe an arc.
3. On the top axis from the R1 Mark , add 1/2 the OC of the insulated small pipe.
4. On the bottom axis from the Inner R1 mark add 1/2 the OC of the insulated large pipe.
5. With dividers set from the apex to the largest dimension, find a common apex and connect the 2 R2 points. Cut out & use as a pattern for the other half.
6. Glue together and apply.

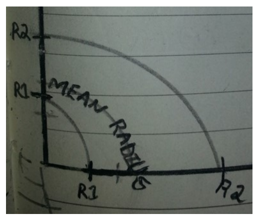


Extruded Foam Reducing Elbows

Concentric

Dims Required: OD & OC of the small & large pipes, insulation thickness.

1. Scribe an arc equal to the mean radius of the large pipe.
2. On the top axis toward the apex mark 1/2 the OD of the small pipe.
3. On the bottom axis toward the apex mark 1/2 the OD of the large pipe.
4. With dividers set to the large OD connect the R1 points by finding a common apex.
5. On the top axis from the Inner (R1) mark, add 1/2 the OC of the insulated small pipe.
6. On the bottom axes from the Inner (R1) mark, add 1/2 the OC of the insulated large pipe.
7. With dividers set from apex to largest dimension find a common apex & connect the R2 points.



Abbreviations

CL	Center Line — - — - — - — -
TOC	Top of Concrete
PSV	Pressure Safety Valve
TCV	Temperature Control Valve
TRC	Temperature Recorder Controller
FRC	Flow recorder Controller
DPI	Delta Pressure Indicator
DPSH	Delta Pressure Suction Head
TIC	Temperature Indicator Controller
LRC	Level Recording Controller
PC	Pressure Controller
TI	Temperature Indicator
FOF	Flush on Face
TOS	Top of Steel
LCV	Level Control Valve
LG	Level Gauge
LC	Level Control
FI	Flow Indicator
FR	Flow Recorder
LA	Level Alarm

PI	Pressure Indicator
T	Steam Trap
FW	Field Weld
BD	Blow Down
PM	Process Module (Piping Mod)
SB	Stick Built

Pipe Designations

P-	401-	3”-	A150
(Process) Line Contents	Line #	Line Size	Flange Size

K-Factor

“**K**” is the thermal conductivity of a material as defined by ASIM (American Society of Testing Materials) as follows:

The amount of heat (BTUs) transmitted through (1) square foot of material (1) inch thick, in (1) hour with a temperature differential of 1°F between the two surfaces of insulation, @ the average or mean temperature of the material.

BTU One (1) BTU is approximately the amount of heat required to raise the temperature of one (1) pound of water.

Mean Temperature:

The artthemtical average between hot and cold faces of insulation.

Example: Pipe temp 200°F, outside temp of insulation 80°F: 200+80 ÷2=140°F (Mean Temp.)

Comparative Insulation Efficiency

Material	K Factor @ 75°F
Foam glass	0.37
Rock Wool (Batts)	0.32
Asbestos Fibre	0.30
Styro Span	0.26
Fibreglass AF 320 3/4 LB	0.26
Fiberglass AF 327 1 LB	0.24
Rigid coated duct	0.24
Fiberglass AF 545	0.23
Fiberglass pipe insulation	0.22
Urethane Foam	0.12 –0.16

The lower the “K” value the better the insulation.

R value = Thickness of Insulation ÷ “K” Factor

Flange Size Chart

Pipe Size	150 #	300 #	400 #	600 #	900 #	1500 #
2	6	6.5	6.5	6.5	8.5	8.5
3	7.5	8.25	8.25	8.25	9.5	10.5
4	9	10	10	10.75	11.5	12.25
5	10	11	11	13	13.75	14.75
6	11	12.5	12.5	14	15	15.5
8	13.5	15	15.5	16.5	18.5	19
10	16	17.5	17.5	20	21.5	23
12	19	20.5	20.5	22	24	26.5
14	21	23	23	23.75	25.25	29.5
16	23.5	25.5	25.5	27	27.75	32.5
18	25	28	28	29.25	31	36
20	27.5	30.5	30.5	32	33.75	38.75
24	32	36	36	37	41	46

All measurements are in inches.

150# is considered a standard and will tolerate up to 150 Lbs. of pressure.

Topic